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Mediating Alchemical Language across Terminologies and Cultures in Ruland's *Lexicon Alchemiae*: A Data-driven Study of Arabic Terms

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From the sixteenth century onwards, reference works emerged as means to transmit technical specialist knowledge to wider audiences. Authors, practitioners, and readers of early modern alchemy and chymistry had to navigate a terminological landscape that drew on Latin, European vernacular languages, and loanwords from languages such as Arabic. Martin Ruland the Younger's *Lexicon Alchemiae* (1612) occupies a central position in the early modern efforts to organise alchemical knowledge through lexicography. Emerging from the Paracelsian tradition yet extending beyond it, the work combines Latin entries with vernacular glosses and incorporates terminology drawn from a wide range of intellectual traditions. Approaching Ruland's *Lexicon Alchemiae* as a reference book, this article investigates how it mediates terminology across linguistic and cultural boundaries, with particular attention to Arabic-derived terms. Drawing on a digital edition, the study combines computational analysis and close reading to examine the structure and contextual use of these terms within the dictionary through selected case studies, such as entries relating to alcohol. The resulting dataset offers new insights into the construction of the lexicon and provides a foundation for further research on Arabic terminology and intercultural knowledge transmission in early modern alchemy.

Introduction

The sixteenth- and seventeenth-century publication boom in technical manuals, reference works, and other knowledge organisation tools saw combinations of diverse forms of knowledges appearing in single reference books. These works not only sold well in the general market but also shaped specialised domains such as alchemy and chymistry as part of an early modern information revolution.¹ Yet while reference works in alchemical contexts have recently received growing scholarly attention, for example through studies of how-to literature or mining treatises, alchemical dictionaries have not yet been examined within this framework.² Martin Ruland the Younger's (1569–1611) *Lexicon Alchemiae* (1612) stands at the crossroads of specialist alchemical terminology and early modern projects of systematising vernacular languages through lexicography.³ Rooted in the Paracelsian tradition yet extending beyond it, the dictionary incorporates older authorities alongside vernacular glosses and foreign-language terms. Ruland's *Lexicon* (Figure 1), thus, represents a culmination of earlier Paracelsian lexicographical traditions while at the same time emancipating itself from this highly specialised context of origin, moving towards a lexicon with broader relevance for the alchemical tradition as a whole. However, Ruland provides almost no information about

¹ Paul M. Dover, *The Information Revolution in Early Modern Europe* (Cambridge: Cambridge University Press, 2021).

² For example Pamela H. Smith, *From Lived Experience to the Written Word: Reconstructing Practical Knowledge in the Early Modern World* (Chicago, IL: University of Chicago Press, 2022); Volkhard Wels, "(Al)Chemisches Wissen im Buchdruck," in *Wissen und Buchgestalt*, ed. Philipp Hegel and Michael Krewet (Wiesbaden: Harrassowitz, 2022), 159–226; or more recently, Tillmann Taape, *Crafting Medicine: Artisans, Knowledge, and the Common Man in Hieronymus Brunschwig's Books on Surgery and Distillation* (Chicago, IL: University of Chicago Press, 2025).

³ On Ruland and his *Lexicon Alchemiae*, see Vladimír Karpenko, "Martin Rulands *Lexicon Alchemiae* im Kontext der chemischen Sprache und Systematik," *Studia Rudolphina* 11, no. 1 (2011): 102–26; Ivo Purš and Josef Smolka, "Martin Ruland the Elder, Martin Ruland the Younger, and the Milieu of the Emperor's Personal Physicians," in *Alchemy and Rudolf II: Exploring the Secrets of Nature in Central Europe in the 16th and 17th Centuries*, ed. Ivo Purš and Vladimír Karpenko (Prague: Artefactum, 2016), 581–605; Thomas Willard, "Hard Places: Paracelsian Neologisms and Early Modern Guides," in *Multilingualism in the Middle Ages and Early Modern Age*, ed. Albrecht Classen (Berlin: De Gruyter, 2016), 355–93; Corinna Gannon, "Martin Ruland d. J., *Lexicon Alchemiae sive dictionarium alchemisticum* (Frankfurt: Zacharias Palthenius, 1612)," in *Matthäus Merian d. Ä. und die Bebilderung der Alchemie um 1600*, ed. Berit Wagner (Frankfurt am Main, Goethe-Universität Frankfurt am Main, 2021), <https://merian-alchemie.ub.uni-frankfurt.de> (accessed 3 May 2026). On early modern alchemical lexicography, see Robert Halleux, "Problèmes de lexicographie alchimiste," in *La lexicographie du latin médiéval et ses rapports avec les recherches actuelles sur la civilisation du Moyen Âge* (Paris, 18–21 octobre 1978) (Paris: Editions du Centre national de la recherche scientifique, 1981), 355–65; Wilhelm Kühlmann, "Materia medica als Problem: Sprachdiskussion und Lexikographie im Paracelsismus," in *Sprachgeschichte und Medizingeschichte: Texte – Termini – Interpretationen*, ed. Jörg Riecke (Berlin: De Gruyter, 2017), 155–82, <https://doi.org/10.1515/9783110524758-011>, Section titled "The historiography of alchemical language," in Sarah Lang, "A Machine Reasoning Algorithm for the Digital Analysis of Alchemical Language and Its Decknamen," *Ambix* 69, no. 1 (2022): 65–83 (on 66–69), <https://doi.org/10.1080/00026980.2022.2038428>. On early modern lexicography more broadly, see Werner Hüllen, "Motives behind Seventeenth-Century Lexicography: A Comparison between German and English Dictionaries of That Time," in *Proceedings of Euralex 1988*, ed. Tamás Magay and J. Zigány, Papers from the EURALEX (Akadémiai Kiadó, Budapest: European Association for Lexicography, 1990), 189–96, <https://euralex.org/category/publications/euralex-1988/> (accessed 3 May 2026); John Considine, *Dictionaries in Early Modern Europe: Lexicography and the Making of Heritage* (Cambridge: Cambridge University Press, 2011); John Considine, "The History of the Concept of Lexicography," in *History of Linguistics 2014: Selected Papers from the 13th International Conference on the History of the Language Sciences (ICHoLS XIII)*, *Vila Real, Portugal, 25–29 August 2014*, ed. Carlos Assunção, Gonçalo Fernandes, and Rolf Kemmler (Amsterdam: John Benjamins, 2016), 31–42. <https://doi.org/10.1075/sihols.126.03con>.

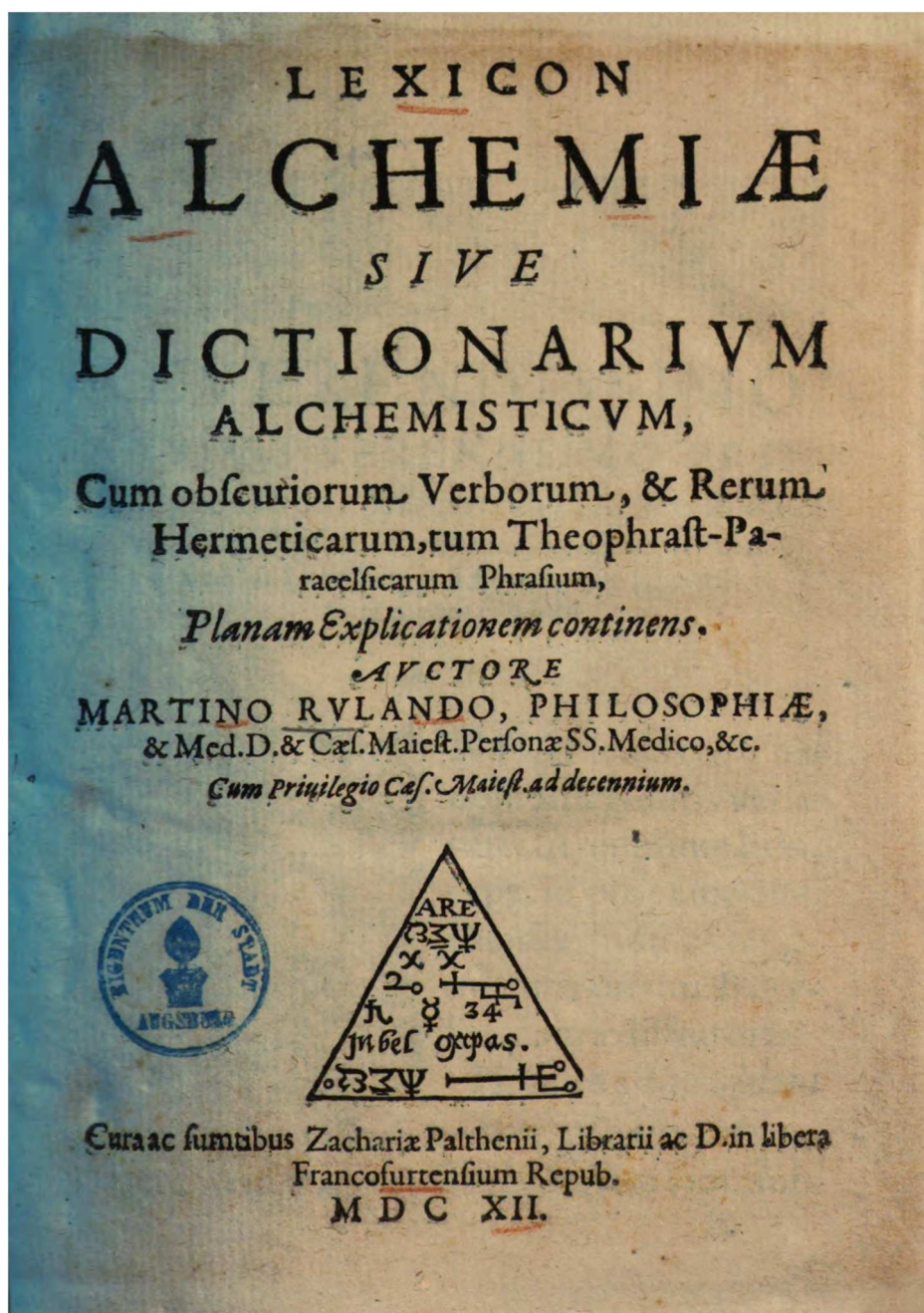


FIGURE 1 Title Page of Ruland's *Lexicon Alchemiae* (1612). Augsburg, Staats- und Stadtbibliothek, 4 Phys 184, title page. Digital copy: BSB-ID 991037514909707356. <https://www.digitale-sammlungen.de/en/view/bsb11221307?page=,1> (15.06.2026)

how the work was compiled. His preface, following the common trope, frames the lexicon as a response to the “Babylonian confusion” of alchemical language, but offers no account of its sources, selection principles, or the intended practical use of the work.⁴

This lack of explicit methodological information leaves scholars to reconstruct Ruland’s working methods from the lexicon itself. The structure of the dictionary is a mixture of brief glosses, synonym lists, and extended conceptual entries and offers potential clues, but does not reveal a consistent organising principle. Rather than a fully systematised reference work, the *Lexicon* often reads as a compiled list of terminological notes. It is precisely in this case that digital methods may offer a promising avenue. A digital edition of the lexicon allows for computational exploration of lexical patterns, repetitions, and overlaps, making it possible to trace connections and sources that are not evident through traditional linear reading.⁵ Specifically, we explore whether recent Large Language Models can help analyse terms pertaining to the Arabic tradition that appear within Ruland in diverse forms and transliterations. A hybrid process of manual and computational engagement illustrates the strength of digital analysis not just in processing scale, but in opening up new modes of reading. As Ruland’s sources remain largely opaque, digital tools can help map the internal structure and external resonances of the lexicon, laying the groundwork for a fuller understanding of its place within early modern alchemical knowledge production and for clues on how this tradition drew on Arabic knowledge and terminology, paving the way for a more global and intercultural history of the alchemical traditions.⁶ Although this study examines a

⁴ In the short, unpaginated preface, Ruland likens the alchemical and Paracelsian traditions to the construction of the Tower of Babel described in Genesis 11. The descendants of Shem, driven by empty glory, rapidly raised a vast structure towards the heavens in the hope of making their names immortal, but were thwarted by the confusion of multiple barbarous languages and forced to abandon their project. A similar fate, he suggests, befalls the alchemists, who are hindered by obscure words and hieroglyphs in their attempts to resolve the relationship between words and things. They seek to extract truth through laborious reading, yet their efforts ultimately prove as futile as those of the builders of Babel. Ruland writes: “In Hermeticis & Theophrasticis libris versaturis, Serenissime Princeps, id propemodum evenit: quod olim Semi filii turrin Babylonicam aedificaturis. Hi enim, dum inani gloria ducti, vastam molem, ad coelos usque erigere, temerario ausu conabantur, immortalitatem nominis inde consecuturi, ab incepto opere linguarum barbaricarum multitudine atque confusione perturbati, turpiter absistere coacti sunt. Illi vero, verborum passim occurrentium obscuritate, & hieroglyphicorum difficultate deteriti, animum partim despondent, artem nobilissimam suique deque habentes: partim pro virili penetrare in verborum rerumque mysteria annitentes, veritatem involucris obtectam densis eruere student, sed tantum lectione laboriosa ista percipientes fructus, quantum gloriae Semi progenies ex laboribus aedificii praeuasti retulerant, oleum & operam perdescentes. Quo utrisque auxilio essem, & Hermeticorum libri non modo diligentius legerentur, sed & intelligerentur, atque sic Ars Alchemiae divina felicius tractaretur: Lexicon hoc Alchemiae olim congestum abs me, iam vero adauctum publicare volui, & S. C. V. dedicatum.” Martin Ruland the Younger, *Lexicon Alchemiae sive dictionarium alchemisticum* [...] (Frankfurt am Main: Zacharias Palthenius, 1612), unpaginated preface.

⁵ Sarah Lang, “Towards a Data-Driven History of Lexicography: Two Alchemical Dictionaries in TEI-XML,” *Journal of Open Humanities Data* 11, no. 20 (2025): 1–6, <https://doi.org/10.5334/johd.303>. Sarah Lang, “Alchemical Dictionaries in TEI-XML,” GitHub repository, <https://github.com/sarahalang/alchemical-dictionaries> (accessed 3 May 2026). Christopher Pollin, “Alchemical-Dictionaries Promptotype,” web application, <https://chpollin.github.io/alchemical-dictionaries/search?q=lapis> (accessed 3 May 2026).

⁶ A recent article has revisited the question of how to name the alchemical tradition, particularly in light of more inclusive and culturally sensitive approaches. While this article continues to refer to the alchemical tradition, it acknowledges and draws attention to that ongoing discussion, see Alexandre M. Roberts, “Disentangling Alchemy,” *Isis* 116, no. 4 (2025): 623–43, <https://doi.org/10.1086/738213>.

lexicographical historical source, the goal of this article is to understand the dictionary as a knowledge organisation resource or reference work. It approaches Ruland's *Lexicon* in the context of broader early modern information management practices. In doing so, it is in dialogue with a second contribution about Ruland's *Lexicon* in this special issue.⁷

After examining how Ruland's *Lexicon Alchemiae* fits within the tradition of Paracelsian lexicography and what is known about the circumstances of its production, this contribution begins with a quantitative analysis based on word counts. This analysis provides insight into structural aspects of the dictionary, for example, by identifying where Arabic-derived terms occur within the work.⁸ The article then interprets these quantitative results in order to explain how the dictionary is constructed. Previous studies have generally approached the lexicon through individual examples of particular terms.⁹ Though somewhat limiting, this approach remains necessary here to some extent, as a dictionary contains such a large number of entries that analysing it in its entirety is difficult. The computational and quantitative analysis therefore serves to provide a broader structural perspective and an efficient heuristic method to detect terms relevant for our study. At the same time, we complement this approach with close readings of selected terms identified in the dataset. In particular, we examine various entries for *alcohol*, amongst other examples. The data generated in this study, thus, provide a basis for future analyses that will examine the Arabic-derived terminology in greater detail. The conclusion

⁷ This paper grew out of discussions that began at a workshop on applying AI to alchemical texts at New York University–Abu Dhabi (NYU-AD) under the title “Alchemy of Global Partnerships.” The workshop's main purpose was to cultivate the potential applications of computational tools, and specifically AI, to multilingual investigations of alchemical texts. Each of the coauthors had been working on different approaches toward Ruland's *Lexicon Alchemiae* unaware of the other's efforts; this was the groundwork for the present joint study. We agreed on the general question of investigating Ruland's sources, and given the well-known influence of Islamic culture on European alchemy, we decided to concentrate on Ruland's Arabic sources. The 2025 Alchemy of Global Partnership workshop v.1 was funded by NYU-AD's Arts+Humanities Research Platform, with additional support from NYU Global Research Initiative, and NYU Dean of Liberal Studies. A second Alchemy of Global Partnership workshop scheduled for 15–18 June 2026 at NYU's Prague campus, is supported by a New York University Global Opportunity Grant, with additional support from NYU Dean of Liberal Studies. The article Vojtěch Kaše and Sarah Lang, “Contextual Word Embeddings for Paracelsian Lexicography: Tangled Terminologies and Their Origins in Ruland's Alchemical Dictionary,” *Ambix*, this issue, should be read in conjunction to this article, as it offers a methodologically different but complementary perspective on the same data.

⁸ This analysis produces a dataset of Arabic terms in Ruland's lexicon, part of which is discussed here, while further aspects will be explored in greater detail in future publications. The dataset as well as code related to the analyses presented here can be found in Sarah Lang, “Arabic-Origin Terms in Ruland's *Lexicon Alchemiae* (1612): Data and Analyses,” GitHub repository, https://github.com/sarahalang/alchemical-dictionaries/tree/main/2026-ambix_arabicTerms (accessed 3 May 2026). The data provide information about the distribution of detected Arabic terms within the dictionary and the contexts in which they appear; Sarah Lang et al., “Confabulated Transliterations? Managing the Lure of Plausibility in LLM-Detected Arabic Terms in an Early Modern Lexicon,” in *Critical Approaches to Automated Text Recognition*, ed. Melissa Terras, et al. (London: Facet Publishing, forthcoming 2026). This article includes more technical discussions and examines the problem of confabulated transliterations in the context of large language models (LLMs) used for automated text recognition (ATR) tasks. In the article, we discuss the potential and limitations of using LLMs in an OCR post-correction and multilingual retrieval workflow to identify Arabic terms that may have been creatively transliterated into early modern Latin. In particular, we examine the lure of plausibility and the difficulty of verifying LLM-generated outputs, especially when dealing with under-resourced or highly ambiguous languages such as Arabic. The article reflects on existing strategies for managing these challenges while demonstrating how LLMs can mislead through confident yet inaccurate suggestions. Therefore we argue for a cautious and critically informed use of LLMs in the humanities.

⁹ See for example, Karpenko, “Martin Rulands *Lexicon Alchemiae*,” or Kühlmann, “Materia medica als Problem.”

situates the *Lexicon* within broader early modern efforts to create reference tools and systems for organising knowledge, reflecting the confluence of different types of knowledge within a single reference work.

Ruland and his Lexicon Alchemiae (1612)

Paracelsian lexicography emerged in the decades following the death of the Swiss medical reformer and alchemist, Paracelsus (1493–1541), as editors and followers sought to clarify his often obscure and newly coined terminology.¹⁰ Beginning in the 1560s, figures such as Adam of Bodenstein (1528–1577), Michael Toxites (1514–1581), and Gerhard Dorn (ca. 1530–ca. 1584) appended glossaries to printed works or issued standalone dictionaries to aid readers navigating Paracelsus’s language.¹¹ These early lexica, primarily in Latin, gradually incorporated multilingual elements. Toxites and Johannes Fischart (1546/1547–1591), for instance, introduced terms in Latin, German, French, and occasionally Italian or Spanish, reflecting the growing appeal of terminological reference books working across linguistic boundaries.¹² Though less expansive and entirely monolingual (Latin only), Dorn’s dictionary established a model of Paracelsian lexicography grounded in philological engagement with Paracelsus’s corpus.¹³ Over time, Paracelsian lexica began to integrate terms that were not strictly Paracelsian, expanding their appeal and function for a broader audience and thus, making their creation more attractive for the book market.¹⁴ Martin Ruland the Younger’s *Lexicon Alchemiae* (1612) is the most comprehensive example of the genre.¹⁵ While grounded in the Paracelsian tradition and drawing on sources such as Dorn’s *Dictionarium*, Ruland significantly expanded its scope.¹⁶ His lexicon contains over 3200 Latin entries across approximately 500 pages, often accompanied by German glosses, ranging from brief statements to

¹⁰ Dietlinde Goltz, “Die Paracelsisten und die Sprache,” *Sudhoffs Archiv* 56 (1972): 337–52; Karl-Heinz Weimann, “Paracelsus-Lexikographie in vier Jahrhunderten,” *Medizinhistorisches Journal* 16 (1981): 167–95; John Considine, “The Beginnings of English Paracelsian Lexicography: Two Collections of Words from Elizabethan Cambridge,” *Ambix* 69, no. 2 (2022): 163–89, <https://doi.org/10.1080/00026980.2021.2012315>.

¹¹ Adam von Bodenstein, *Onomasticum*, in *Großenn Wundartzney* [...] (Straßburg/ Frankfurt am Main, Adam von Bodenstein, 1566); Johann Fischart and Michael Toxites, *Onomasticum alterum, Theophrasti Paracelsi* (Strasbourg: Jobin, 1574); Johann Fischart and Michael Toxites, *Onomasticon Theophrasti Paracelsi*, Noscemus Wiki, last modified 7 October 2021, http://wiki.uibk.ac.at/noscemus/Onomasticon_Theophrasti_Paracelsi (accessed 3 May 2026); Gerhard Dorn, *Clavis totius Philosophiae Chymisticae* (Lyon: Apud haeredes Iacobi Iunctae, 1567), <https://emlap.flu.cas.cz/wiki/book/21> (accessed 22 February 2026); Gerhard Dorn, *Paracelsi Dictionarium*, in *Fasciculus Paracelsicae Medicinae* (Frankfurt am Main: Johannes Spies, 1581), <https://emlap.flu.cas.cz/wiki/book/75> (accessed 22 February 2026); Gerhard Dorn, *Dictionarium Theophrasti Paracelsi* (Frankfurt am Main: Christoph Corvinus, 1583); Gerhard Dorn, “Dictionarium Theophrasti Paracelsi,” Noscemus Wiki, last modified 18 February 2022, http://wiki.uibk.ac.at/noscemus/Dictionarium_Theophrasti_Paracelsi (accessed 3 May 2026).

¹² Fischart and Toxites, *Onomasticum alterum*.

¹³ Dorn, *Clavis totius Philosophiae Chymisticae*; Dorn, *Paracelsi Dictionarium*; Dorn, *Dictionarium Theophrasti Paracelsi*.

¹⁴ For more information on the Paracelsian lexicographical tradition, see Vojtěch Kaše and Sarah Lang, “Contextual Word Embeddings for Paracelsian Lexicography: Tangled Terminologies and their Origins in Ruland’s Alchemical Dictionary,” *Ambix*, this issue.

¹⁵ Some earlier literature mistakenly attributed the *Lexicon* to Ruland the Elder, but this is incorrect: the father neither authored nor co-authored the work, although the son undoubtedly drew upon his father’s extensive medical and alchemical knowledge. Karpenko, “Martin Rulands *Lexicon Alchemiae*,” 102–03.

¹⁶ Karpenko, “Martin Rulands *Lexicon Alchemiae*,” 118–19.

extended articles, including synonym lists and conceptual discussions.¹⁷ While Latin is the dominant language, German translations and paraphrases are frequent, and the *Lexicon* includes a notable number of foreign-language terms, including in Arabic, which shall be investigated later in more detail.

Publication circumstances

Having been appointed in 1607 and ennobled by the end of 1608, Ruland was closely involved in the Emperor Rudolf II's (1552–1612) alchemical and medical investigations. Unlike his father, Martin Ruland the Elder, who also worked in Rudolf's court and pursued wider scholarly interests, Martin Ruland the Younger dedicated his career entirely to Paracelsian chemiatry. Although Ruland the Younger did not equal his father in the number or range of his writings, he left a far more enduring mark on the history of science through his *Lexicon Alchemiae*. Compiled in Prague during his appointment to Rudolf II, the *Lexicon* was published posthumously and dedicated to Heinrich Julius of Brunswick (1564–1613), the Duke and chairman of the Emperor's privy council during the last years of Rudolf's reign, who was, himself, deeply engaged in medicinal alchemy, which Ruland acknowledges in his dedication. The imperial privilege against unauthorised reprinting for ten years suggests that it was expected to have commercial success.

Despite its significance, the dictionary remains understudied as a whole, offering ample material for future research, particularly in light of its multilingual richness.¹⁸ Most modern references cite individual entries rather than analysing the work itself.¹⁹ Ruland's preface offers little detail about his sources or methodology, and his citations are sparse, typically limited to names of older authorities rather than specific texts or chapters. As a result, much about the *Lexicon's* compilation principles remains unclear and must be inferred from the structure and content of the entries themselves. A mundane circumstance may have influenced this final form of Ruland's *Lexicon*. The aforementioned imperial printing privilege for the *Lexicon Alchemiae*, issued on 2 September 1607, indicates that Ruland had been working on the lexicon since his admission to the imperial court on 24 May 1607 and which may well have been one of his duties as an imperial servant.²⁰ Martin Ruland the Younger died of the plague in Prague on 23 April 1611. His

¹⁷ Martin Ruland the Younger, *Lexicon Alchemiae sive dictionarium alchemisticum* [...] (Frankfurt am Main: Zacharias Palthenius, 1612); Arthur Edward Waite, trans., *Martinus Rulandus: A Lexicon of Alchemy, or Alchemical Dictionary Containing a Full and Plain Explanation of All Obscure Words, Hermetic Subjects, and Arcane Phrases of Paracelsus* (London: John M. Watkins, 1964); Martinus Rulandus (Martin Ruland the Younger), "Lexicon Alchemiae," Nosce Wiki, last modified 27 May 2021, http://wiki.uibk.ac.at/nosceWiki/Lexicon_Alchemiae (accessed 3 May 2026).

¹⁸ Purš and Smolka give an excellent overview over what is known of Ruland's career, yet lament that both Rulands, father and son (the Elder and the Younger) lack systematic treatment in the form of dedicated scholarly monograph, see Purš and Smolka, "Martin Ruland the Elder, Martin Ruland the Younger," 584–85.

¹⁹ An example can be found in: Carmen Schmechel, "Leaven of Dough, Ferment of Gold: The Breadmaking Analogy in Medieval Metallic Transmutation," *Ambix* 71, no. 3 (2024): 243–70, <https://doi.org/10.1080/00026980.2024.2396702>.

²⁰ Purš and Smolka, "Martin Ruland the Elder, Martin Ruland the Younger," 592.

dedication to Heinrich Julius of Brunswick, dated 10 April 1611 in Prague, fits this timeline and suggests that he likely wrote this prefatory material himself rather than it being added later to an almost completed work. However, his publisher, Zacharias Palthenius (1579–1615) in Frankfurt, did not publish the book until the following year. It is therefore possible that Ruland did not complete the final editing of the work before his premature death. The absence of an errata page indicates that Ruland likely never reviewed the final printed version.²¹ His untimely death may thus have resulted in the *Lexicon* being printed in a state that was not fully completed or revised. Unless new archival sources emerge that shed light on the circumstances, questions around the state in which the manuscript was left, whether Ruland considered it complete or intended to revise and further systematise it, how it reached the publisher, and whether additional edits were made there, cannot be answered with any certainty. The recent creation of a digitally encoded version offers the opportunity to explore the work through computational methods, which may help reconstruct its sources, analyse its use of Arabic terminology and reveal new dimensions of its organisation.²²

What's in a name? Ruland's title and the tradition of glossaries

Ruland's work is not merely a linguistic dictionary but occupies a transitional space within the evolving tradition of alchemical knowledge systems as part glossary, part dictionary and part encyclopaedic reference.²³ While these terms are similar and often used interchangeably in everyday language, they have distinct meanings in modern usage. A *dictionary* describes words from a linguistic perspective, whereas an *encyclopaedia* collects knowledge about subjects, usually organised under alphabetical headwords. In linguistics, a *lexicon* refers to the vocabulary of a language or of a particular field of knowledge, deriving from the Greek term for “word.” In the context of reference works, however, the term is used somewhat differently. Although it retains the principle of alphabetical organisation, an encyclopaedic dictionary typically contains longer entries that provide more detailed information about the subjects treated. By contrast, a dictionary in the modern sense is organised primarily according to linguistic concerns and historically developed from word lists and glossaries. It is important to note, however, that the encyclopaedia acquired its modern meaning only during the Enlightenment. As a result, this

²¹ We are grateful to the reviewers for this important comment.

²² Lang, “Towards a Data-Driven History of Lexicography”; Lang, “Alchemical Dictionaries in TEI-XML”; Pollin, “Alchemical-Dictionaries Promptotype.”

²³ For more discussion of those topics, see Sarah Lang, “Digitale Erschließungsmethoden für alchemische Texte am Beispiel der *Symbola Aureae Mensae* Michael Maiers,” in *Alchemie – Genealogie und Terminologie: Bilder, Techniken und Artefakte*, ed. Petra Feuerstein-Herz and Ute Frietsch, Wolfenbütteler Forschungen 166 (Göttingen: Harrassowitz, 2022), 133–52; Sarah Lang, “Vom ‘Wissen in Buchform’ zum formalen Wissensmodell: Digitale Aufbereitung wissenschaftshistorischer Drucke am Beispiel der *Alchemica* Michael Maiers,” in *Wissen und Buchgestalt*, ed. Philipp Hegel and Michael Krewet (Wiesbaden: Harrassowitz, 2022); Petr Pavlas, “From Circle to Book: The Evolution of Metaphors and the Birth of Early Modern Encyclopaedism,” in “Cognitive Metaphors and Encyclopaedic Knowledge: Exploring Semantic Transformations in Early Modernity,” ed. Petr Pavlas, Lenka Řezníková, and Lucie Storchová, *Filosofický časopis* 73, no. 1 (2025): 35–55, <https://doi.org/10.46854/fc.2025.1s.35>.

category did not exist in its present form in Ruland's time. Ruland's *Lexicon* therefore resembles an encyclopaedia in some respects, while also retaining features of a word list or glossary, making it a hybrid form. A *glossary* or *clavis*, by contrast, usually refers to a list of specialised terms relating to a particular domain and is often appended to a specific work to which it serves as a guide, as can be seen in the Paracelsian lexicographical tradition. The term *lexicon*, especially in the German context, can refer more generally to a reference work and is often further specified to indicate its type, for example as a *realia lexicon*. However, this broader meaning became clearly established only in the eighteenth century. Earlier titles such as *Lexicon sive dictionarium* that use the terms as synonyms indicate that the term had not yet fully moved away from its origins in describing the words of a language in an alphabetically organised manner. This ambiguity is evident in Ruland's title: *LEXICON ALCHEMIAE SIVE DICTIONARIUM ALCHEMISTICVM, Cum obscuriorum Verborum, & Rerum Hermeticarum, tum Theophrast-Paracelsicarum Phrasium, Planam Explicationem continens*. Here the work is described as a *lexicon* or *dictionary* that provides clear explanations not only of obscure words but also of things and, specifically, Paracelsian phrases.²⁴ As the overview of Paracelsian lexicography above has shown, titles such as *Dictionarium* or *Clavis* were more common choices than *Lexicon*.

According to Vladimír Karpenko, Ruland's treatment of terminology not only reflects the *Lexicon*'s intellectual ambition and Paracelsian influence but also draws on older classificatory systems such as those of Rhazes.²⁵ Ruland's dictionary builds on Dorn's much shorter *Dictionarium*, and in some entries mirrors it closely, but it also departs from it in structure and content.²⁶ While earlier lexica were often appended to editions of Paracelsus's works, Ruland's *Lexicon* is an independent reference text, similar to some Paracelsian standalone lexica before him. It not only addresses material substances but also abstract concepts such as *Alchemia* and *Chemia*, thus situating it between a linguistic dictionary, encyclopaedia, and thesaurus. By Ruland's time, numerous printed and manuscript glossaries were available, and he could have drawn on either tradition.²⁷ It is thus not a unique characteristic of Paracelsian lexicography that such texts were situated between practical tools for explaining specific texts and forming part of a broader tradition of compiling glossaries aimed at creating repositories of knowledge.

²⁴ The duality of the title as *Lexicon* and *Dictionarium* may further be understood as an example of the stylistic device known as *hendiadyoin* or *hendiadys* (literally, "expressing one thing through two"), frequently used in Latin to indicate something possessing aspects of both terms while also amplifying the meaning. Well-known examples include *vi et armis* ("by force and arms," i.e. "by armed force"), the Shakespearean "sound and fury" for "furious sound," or the modern English expression "nice and cool." Selecting only one of these terms in the present discussion might create the impression of an anachronistic conceptual clarity. For this reason, the terms are used interchangeably here, both for stylistic variation and to emphasise different aspects of the work under discussion.

²⁵ Karpenko, "Martin Rulands Lexicon Alchemiae," 108–11.

²⁶ Karpenko, "Martin Rulands Lexicon Alchemiae," 118–19.

²⁷ Sébastien Moureau, "Azoc id est mercurius. Les glossaires manuscrits alchimiques," *Micrologus: Natura, Scienza e Società Medievali / Nature, Science and Medieval Societies* 34 (2026): 235–75, <https://hdl.handle.net/2078.5/272649> (accessed 15 June 2026).

Structural analysis: visualising Ruland's Lexicon Alchemiae

To better understand the structure and content of Martin Ruland's *Lexicon Alchemiae* (1612), we will now approach the dictionary through its quantitative makeup by probing its structural features such as entry word length. Utilising the structured nature of the dataset available to us and drawing on our detection of Arabic terms to be discussed later, these visualisations offer insights into both the overall composition of the dictionary and the distribution of Arabic (or Arabic tradition) terms within it: This study can build on the availability of a digital version of Ruland's *Lexicon*, encoded using the Text Encoding Initiative (TEI) dictionary module. TEI encoding enables systematic examination of individual entries independently of their spatial placement within the source documents and supports the automatic comparison of identical lemmas across multiple lexicographical sources. Such structured digital formats enable macro-level investigations into early modern terminology and the conceptual organisation of knowledge, as we demonstrate in this article. This encoding allows efficient retrieval of, for example, all lemmas beginning with a specific letter, such as A, since this information is encoded in a structured and machine-readable way.²⁸ The dictionary contains approximately 3200 entries, and through this encoding it became possible to access each entry and its corresponding headword directly, regardless of its position on the page. Even entries spanning two pages are captured in full. This approach prevents the high error rate that would result from searching through unstructured PDFs or plain transcriptions. Having the entries clearly segmented and addressable allows us to explore the dictionary through statistics.

While the one part of our study consists of a strictly quantitative analysis based on a fully human-annotated and curated dataset, structured according to data standards common in the digital humanities, the next builds on a new derived dataset built specifically for this article using a workflow supported by Large Language Models (LLMs).²⁹ Although the limitations of LLMs are well documented, their

²⁸ The underlying data originates from Handwritten Text Recognition based on NOSCEMUS GM high-performing general model, which produced the initial transcription. This transcription was annotated semi-automatically as TEI-XML, using simple search and replace functions to accelerate the process. The digital data are published and explained in: Lang, "Towards a Data-Driven History of Lexicography." Lang, "Alchemical Dictionaries in TEI-XML." A web application allows users to explore the data through simple queries, providing not only a version of Ruland's dictionary that is queryable without programming skills but also a first impression of data quality: Pollin, "Alchemical-Dictionaries Promptotype." However, since the analyses discussed in this article were achieved through programming-enabled, direct manipulation of the underlying dataset, we were able to be more flexible and precise in the questions we asked and thus obtained more results than would be possible through the web interface alone.

²⁹ We initially experimented with different LLMs, such as OpenAI's ChatGPT 4o or Gemini Flash, to see which performed best and whether there were significant differences in performance. However, we found no drastic differences in performance and ultimately ran all our analyses using OpenAI's ChatGPT 4o (in a period between August and December 2025). The technical workflow and performance evaluation is detailed in a dedicated publication, where we also address methodological challenges, including those related specifically to lower-resource languages such as Arabic. See Lang et al., "Confabulated Transliterations?" This chapter explains in more detail exactly what was done, and how it was done, and it also problematises the approach, highlighting pitfalls, especially when working with lower-resource languages such as Arabic. As most LLMs are predominantly trained on English data and thus perform best in that language, performance can be less reliable in underrepresented languages. As many users of these tools will already know, even in English, the outputs are not always fully trustworthy. The

effectiveness is strongly contingent upon the prompting strategies employed. More explicit and constrained prompts significantly reduce the likelihood of hallucinated outputs, as opposed to open-ended queries. In our case, no LLM-generated data were fed directly into the final analytical outputs or visualisations. Instead, the model produced a structured table of candidate terms, enriched with relevant meta-data to assist subsequent human verification. This table formed the basis of a multi-stage human-in-the-loop validation process in which several researchers checked the accuracy and plausibility of each entry.³⁰ Our methodological priority in the detection phase was to reduce the risk of missing relevant terms, even if this meant accepting a higher number of false positives, which resulted in the inclusion of many terms later identified as Latin rather than Arabic. The result was an initial list of approximately 1000 entries, which was subsequently reduced through human review to around 400 confirmed or probable Arabic terms.³¹ While this implies a high rate of exclusion, the retained data included many terms that are well attested in the Arabic alchemical tradition, suggesting that material relevant to alchemical traditions is indeed represented in the LLM's training corpus. Some excluded terms were clearly misidentified, including obvious Latin entries; others were more ambiguous and required further interpretation. Nevertheless, the final list represents a significant dataset of historically relevant terms.

The creation of this derived dataset using the above method was necessitated by limitations in the structured data available for our analysis. For instance, while the available data encoded following the *Text Encoding Initiative* standard (TEI-XML) allowed us to assess properties such as entry length or distribution, foreign-language terms in languages other than German had not been annotated in the dataset. Manual identification of such terms across the corpus would have been prohibitively time-consuming. By contrast, the LLM was able to generate a preliminary set of candidate terms rapidly, writing the results into a table which included ample contextual information to support manual review, including retrieval-augmented generation (RAG) elements to enhance the quality of suggestions prior to human validation. Only after this cleaning process were the final visualisations produced.

The principal advantage of this workflow is twofold. First, it significantly reduced the manual labour that would otherwise have been required to extract foreign-language terms from an unannotated corpus. Second, it overcame the limitations of

²⁹ *Continued*

authors of this study are, of course, fully aware of this fact and took this knowledge into account when designing the workflow described here. However, effective prompting strategies have been shown to mitigate this. All relevant code repositories, data and prompts are included in the GitHub repository linked in footnote 8. On LLMs see also Sarah Lang, "Critical Concerns for Using LLMs in the (Computational) Humanities and Beyond," in *Large Language Models for the History, Philosophy, and Sociology of Science: Reflections from a Field in Motion*, ed. Arno Simons et al. (Bielefeld: transcript, 2026), 35–50, <https://www.transcript-verlag.de/978-3-8376-7994-6/understanding-science-with-large-language-models/?c=313000027> (accessed 3 May 2026).

³⁰ See Eduardo Mosqueira-Rey et al., "Human-in-the-Loop Machine Learning: A State of the Art," *Artificial Intelligence Review* 56, no. 4 (2023): 3005–54, <https://doi.org/10.1007/s10462-022-10246-w>.

³¹ Our table is included as a supplementary file for review and has been published in a code repository linked in footnote 8.

traditional search-based approaches which presuppose prior knowledge of the precise terms one is attempting to locate and depend on near-exact spelling. In the case of early modern adaptations of Arabic terms, many of which due to the inexistence of standardisation appear in countless variants, often compounded by OCR artefacts, this would have severely constrained recall. The LLM, by contrast, proved capable of recognising and grouping highly divergent spellings of known terms, thereby identifying instances that traditional string-matching would likely have missed. Because LLMs have been trained on data containing spelling variations, errors, and OCR artefacts, they can detect terms even when they do not precisely match expected forms. Moreover, their operation includes a semantic component, rather than relying solely on letter-by-letter matching, which allows them to spot even heavily corrupted word forms. This capacity to detect irregular and historical renderings of Arabic terms marks a key advantage over previous techniques and demonstrates the utility of the LLM-based approach within this specific historical-linguistic context.

Dictionary and entry structure

The first bar chart graphic produced through the aforementioned workflow (Figure 2) groups all entries by word count. Of the approximately 3 200 entries in the *Lexicon*, nearly 1900 are fewer than nine words long: essentially short, glossary-style definitions (e.g. Figures 5–7) rather than extended discussions. About 550 entries fall into the 10–24 word range, 250 between 25–49 words, and around 125 entries exceed 200 words, with the longest approaching 300. The next graphic (Figure 3) offers a second view of this aspect by plotting entry length. With entries levelling off after approximately 350 words, it confirms that most entries are comparatively short. Yet, as longer entries occupy more space, they still

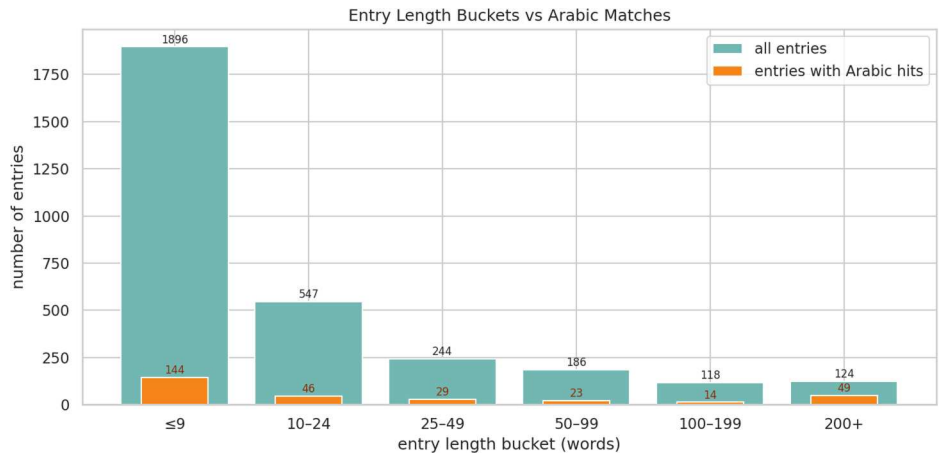


FIGURE 2 The entry bucket length groups entries by word count, showing that the vast majority, including most with detected Arabic terms, are short, containing fewer than nine words. The maximum length is approximately 300 words.

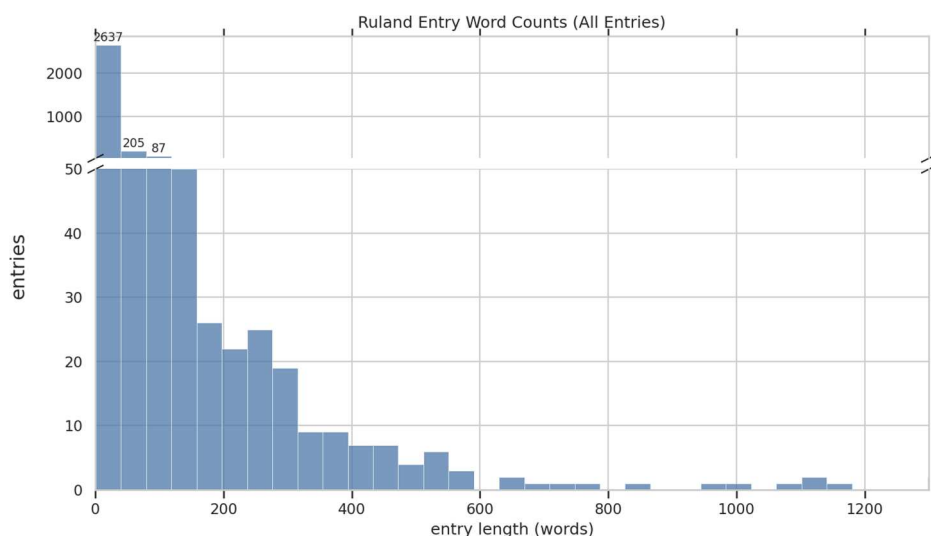


FIGURE 3 This bar chart plots the entry length in words, showing that there are a few outliers with more than 1000 words. However, as the left side of the chart clearly shows, the bulk of entries is up to 200 words long, with 2637 entries being below forty words in length.

constitute a significant portion of the lexicon's overall text and page count. These observations on entry lengths are significant because they clearly demonstrate that Ruland's *Lexicon* differs markedly from earlier Paracelsian dictionaries. The content of these earlier works falls almost entirely within the category of entries shorter than nine words, whereas Ruland's *Lexicon* also contains a significant number of substantially longer entries.³²

A recent survey of Latin alchemical manuscript glossaries from the fourteenth to seventeenth centuries shows that many already employed formats familiar from Ruland's *Lexicon*, including synonym formulae of the type "A id est B," grouped lists of equivalents and short definitions. Extended discursive entries, however, are comparatively uncommon.³³ That Ruland's longer entries sometimes provide extended explanations, indicate the source of particular meanings, or situate them within a broader conceptual context suggests that Ruland introduced a more expansive form of lexicographical practice, approaching an encyclopedic format. In this way, Ruland's lexicon offers a considerably more comprehensive treatment of alchemical terminology than its predecessors.

Multilinguality and Arabic terms within the *Lexicon*

Ruland's *Lexicon* operates within the Latin–German tradition that emerged from Paracelsian lexicography. Its compilers had to navigate Paracelsus' preference for

³² In fact, even the nine-word category is too generous for those earlier dictionaries, as most of their entries contain fewer than five words.

³³ Moureau, "Azoc id est mercurius," *Micrologus: Natura, Scienza e Società Medievali / Nature, Science and Medieval Societies* 34 (2026): 235–75, <https://hdl.handle.net/2078.5/272649> (accessed 15 June 2026).

German (with the resulting proliferation of a range of specialised or newly coined Paracelsian terms) alongside the lasting influence and far greater scholarly prestige of Latin.³⁴ This tension is visible in Dorn's efforts to elevate the status of Paracelsus' works by rendering them in Latin. Consequently, Paracelsian dictionaries, including Ruland's, frequently shifted between Latin and German, promoting the increased vernacular use of pharmaceutical and alchemical terminology, as well as the re-Latinisation of certain German terms, which can equally be observed in the efforts to Latinise German mining terminology.³⁵ This linked Paracelsian, alchemical, and medical vocabularies in a shared professional space that included less erudite practitioners engaging with increasingly multilingual knowledge sources. Navigating multilingualism thus became a defining feature of this intellectual environment.³⁶ In his recent survey of Latin alchemical manuscript glossaries from the fourteenth to seventeenth centuries, Sébastien Moureau shows that among non-Latin foreign-language expressions, Arabic terms form the largest category.³⁷ The prominence of Arabic-derived terms is due to the centrality of Arabic sources in the formation of Latin alchemical literature. He emphasises that their Latinised forms are often heavily distorted and sometimes so corrupted as to be difficult or impossible to identify. Some may even represent accidental or deliberate fabrications rather than genuine borrowings from Arabic.³⁸

Such distortions may have arisen through multiple stages of transmission, including mishearings during translation or misspellings from repeated copying over long periods. Indeed, by the early modern period, some of these terms had acquired meanings independent of their linguistic origins. The Paracelsian movement intensified interest in specialised and foreign terminology and much like Egyptian hieroglyphs were interpreted without knowledge of the language itself in early modern emblematic traditions, Arabic-derived terms could become vehicles for new meanings generated within alchemical tradition, functioning like *Decknamen*.

But why did Arabic-derived terms appear to occupy such a distinctive status within alchemical lexicographical tools? As Charles Burnett argues, the success of the medieval translation movement rested on confidence that a faithful literal translation preserved the intellectual content of Greek and Arabic science, achieving a *translatio studii* whereby knowledge was transferred from one culture to another.³⁹ By the thirteenth century Latin translations were widely regarded as

³⁴ Goltz, "Die Paracelsisten und die Sprache."

³⁵ Dominik Berrrens, "Names and Things: Latin and German Mining Terminology in Georgius Agricola's *Bermannus*," *Antike und Abendland* 65–66, no. 1 (2020): 232–43, <https://doi.org/10.1515/anab-2019-0011>.

³⁶ On multilinguality and plurilingualism, see Glenn W. Most, Dagmar Schäfer, and Mårten Söderblom Saarela, eds., *Plurilingualism in Traditional Eurasian Scholarship: Thinking in Many Tongues*, Ancient Languages and Civilizations 3 (Berlin: De Gruyter, 2023).

³⁷ Moureau, "Azoc id est mercurius," 249.

³⁸ Moureau, "Azoc id est mercurius," 251–52.

³⁹ Charles Burnett, "The Translator as an Authority," in *Translation and Authority – Authorities in Translation*, ed. Pieter De Leemans and Michèle Goyens, The Medieval Translator (Turnhout: Brepols, 2017), 53–67, <https://doi.org/10.1484/M.TMT-EB.5.109400>; Charles Burnett, "Translation and Transmission of Greek and Islamic Science to Latin Christendom," in *The Cambridge History of Science, vol. 2: Medieval Science*, ed. David

sufficient substitutes for the originals. Although Renaissance humanism encouraged renewed engagement with original-language sources, especially in Greek, the medieval ideal of *translatio studii* remains important for understanding Arabic-derived terminology in alchemical texts. Such terms may have entered Latin through direct translation from Arabic manuscripts, oral transmission, or mediation by intermediaries from a range of backgrounds. Technical glossaries formed part of this process, helping readers navigate terms that were only transliterated rather than translated into Latin.

Ruland rarely cites his sources. When he does, he typically names classical authorities such as Pliny the Elder (23/24–79) or Avicenna (Ibn Sina, ca. 980–1037), and only occasionally gives book and chapter references. In Arabic-derived entries, terms are sometimes introduced with phrases like “the Arabs call it,” but without further specification of where this information would have come from. It is unclear whether Ruland had direct access to Arabic texts or relied on Latinised Arabic traditions, such as the works of Avicenna or Geber Latinus (fl. late thirteenth to early fourteenth century).⁴⁰ While his use of Arabic terms underscores the intercultural nature of alchemical knowledge, it also raises questions about how early modern scholars interpreted and adapted non-European sources.

Returning to our analysis, overlaying the data on entry lengths discussed earlier with the frequency of Arabic terms reveals that such terms often appear in short entries (Figure 2), typically in concise formulations like “X means Y” without further elaboration. Yet, looking into the relative statistics, it becomes clear that longer entries, where one might anticipate more frequent use of multilingual references or historical elaboration, do indeed show a proportionally higher presence of Arabic. When we compute the *relative* share of Arabic-containing entries across length categories (in Figure 2), a clear trend emerges: while Arabic terms appear most frequently in *absolute* numbers within the shortest entries under nine words, they become increasingly *proportionally* frequent in longer entries. Entries in the “below 9” and “10–24” word categories contain Arabic terms at rates of 7–8%, rising to 12% in the “25–49,” “50–99” and “100–199” categories. This trend continues, reaching its peak at 37% among entries over 200 words. Though fewer in number, longer entries are thus significantly more likely to include Arabic terms, reflecting their greater tendency to offer extended discussions including multilingual explanations. Consequently, Arabic terms are not limited to learned digressions but are integrated into the basic vocabulary of Ruland’s *Lexicon* as well. All in all, the above findings regarding entry lengths clearly reflect the hybrid nature of Ruland’s *Lexicon*, at the intersection

³⁹ *Continued*

C. Lindberg and Michael H. Shank (Cambridge: Cambridge University Press, 2013), 341–64, <https://doi.org/10.1017/CHO9780511974007.016>.

⁴⁰ The author of Ps.-Geber texts is most likely the Italian Franciscan monk Paul of Taranto, see William R. Newman, *The Summa Perfectionis of Pseudo-Geber: A Critical Edition, Translation, and Study* (Leiden: Brill, 1991).

of glossary-like “hard word lists” but moving towards more encyclopedia-like forms in the longer entries, witnessing the beginnings of a lexicon format in the modern sense.⁴¹

Navigating the *Lexicon* and its contents

The dictionary is alphabetised and this reflects how a user would likely encounter its terms. Unsurprisingly, the letter A dominates in potential Arabic terms identified (Figure 4). This is not surprising as the Arabic definite article *al-* appears in many loanwords. However, the A-section of Ruland’s dictionary also contains by far the most entries overall. The concentration of many Arabic terms in concise definitions and their distribution across the alphabet suggest that while Ruland’s work was grounded in Latin–German lexicography, it also drew on a wider multilingual and intercultural knowledge tradition. Positioned between glossary and encyclopaedia, the *Lexicon Alchemiae* thus reflects both the practical needs and scholarly ambitions of early seventeenth-century alchemical writing. Indeed, as a hybrid work – part glossary, part encyclopaedia, part compendium of practical knowledge – the *Lexicon Alchemiae* reflects the tension between Ruland’s stated desire to impose order on the disorderly language of alchemy and the realities of early modern compilation. Although Ruland declares in his preface that the lexicon is intended to resolve the chaos of alchemical terminology, the internal organisation of the work remains only partially systematised.

As we have established earlier, entries vary dramatically in length but they also vary in scope. Many are short, glossary-like definitions consisting of little more than a headword and brief Latin gloss, sometimes with German or even multilingual synonyms (see Figure 5). Others, such as the entries for *Sal* or *Lapis*, span several pages and include conceptual elaboration, terminological comparisons across languages and occasionally even practical recipes. These longer entries suggest Ruland’s ambition not only to define primarily Paracelsian words but also alchemical words, processes and ideas more generally. It is especially telling that one of the most elaborate entries (*lapis*) falls outside the core Paracelsian tradition. Its inclusion illustrates how Ruland extends beyond purely Paracelsian lexis, and also suggests that he may have felt the need to compensate for the absence of such material in his sources by constructing a particularly detailed entry of his own, likely drawing on a range of written and oral sources. According to Wilhelm Kühlmann, Ruland thereby opens up the schema of older *Onomastica*, transforming it into an alphabetised *realia* lexicon that integrates terminology

⁴¹ For a comparative perspective on early modern lexicographical traditions, see Werner Hüllen’s work on seventeenth-century Latin–German and Latin–English dictionaries. He shows that German lexicography was oriented toward linguistic systematisation and the elevation of the vernacular, whereas English “hard word” lists served more practical aims, explaining Latinate terms for less-educated readers. Hüllen, “Motives behind Seventeenth-Century Lexicography.”

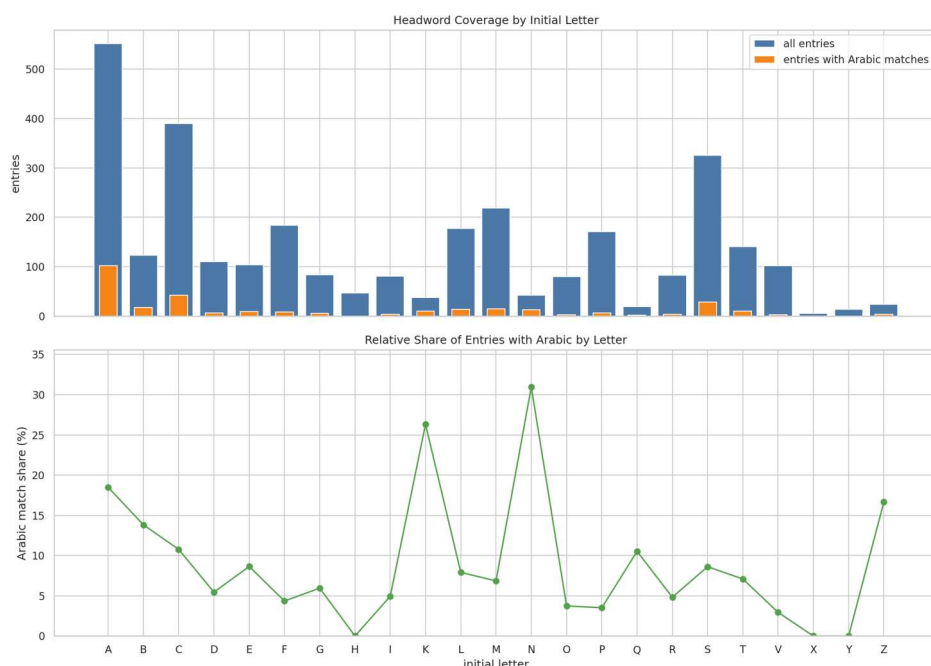


FIGURE 4 This bar chart plots the number of lemmata (headwords) or entries by starting letter, including the number of entries containing Arabic terms per letter. Unsurprisingly, A is by far the most dominant. However, as the top graphic shows, A also contains the highest number of entries overall, regardless of Arabic term detection.

from a broad range of sister disciplines such as metallurgy, thereby moving far beyond Paracelsus' original medico-chemical focus.⁴²

Several entries in Ruland's *Lexicon* are unusually long, some even make use of list structures or Ramist diagrams.⁴³ The *Lapis* entries alone span roughly thirty-five pages, while the *Sal* entries cover around eleven pages. Yet, despite the centrality of the term and its many variations in the alchemical tradition, all *Mercurius* entries are confined to only four pages, making up a comparatively small portion of the dictionary. The entry on *Naphtha*, for instance, extends over nearly three pages and discusses bituminous shale oil deposits, citing classical authors such as Dioscorides and Pliny the Elder. By contrast, Soukup remarks that the entry *Mercurius Solis* (on 333) is surprisingly brief, barely acknowledging the concept's

⁴² Kühlmann, "Materia medica als Problem." Originating in medieval Latin, *realia* refers to expressions denoting "real," that is, material objects or things, rather than abstract concepts. *Onomastics* (or *onomastica lexis*) refers to the linguistic origins, meanings, and etymology of proper names.

⁴³ "Lapis" (all entries), Ruland, *Lexicon Alchemiae*, 273–308; "Mercurius" (all entries), Ruland, *Lexicon Alchemiae*, 331–35; "Mercurius Solis," Ruland, *Lexicon Alchemiae*, 333; "Naphtha," Ruland, *Lexicon Alchemiae*, 342–44; "Sal" (all entries): Ruland, *Lexicon Alchemiae*, 410–21; "Nitrum," Ruland, *Lexicon Alchemiae*, 345–48 includes a Ramist diagram. Of the *Lapis* entries, only pages 292–300 refer to the Philosophers' Stone, where Ruland constructs a recipe, drawing on a range of authors, many of whom are cited by name, and in some cases even by folio. This level of citation and the construction of a recipe are not the norm within the *Lexicon*, making this section stand out from much of the rest of the dictionary.

existence, despite its known significance in the Emperor's laboratory.⁴⁴ This suggests that the *Lexicon* does not directly reflect the specific interests of the Prague court either.

Investigating the headwords relating to Ruland's Arabic terms

The first entry in Ruland's *Lexicon Alchemiae*, dedicated to lead (*plumbum*), listed under the headwords of Abam, Abartamen, Aiarazat and Accib (see Figure 5), already signals the structural and linguistic complexities that typify the work. It contains Latin and German synonyms (e.g. *Bley*), as well as a broad array of alchemical *Decknamen*, establishing a precedent for the lexicon's polyglot and terminologically dense style. Throughout the volume, Ruland addresses materials, processes, and concepts across multiple languages, as mentioned already, often without consolidating variants under a single headword. This is particularly evident in the case of Arabic-derived terms, which frequently appear under inconsistent spellings, distributed irregularly across the text. Such editorial choices demonstrate Ruland's adherence to an unsynthesised, source-driven compilation strategy. Rather than integrating variant forms or providing interpretive glosses, he maintains discrete entries, even when the conceptual content overlaps. This results in multiple listings for essentially the same term, distinguished only by minor orthographic variation. Ruland's handling of the term *alcohol* in the *Lexicon Alchemiae* exemplifies the orthographic and structural inconsistencies that characterise the work as a whole. *Alcohol* appears under a range of variant spellings, divided between two distinct alphabetic placements ('ac' and 'ak').⁴⁵ Such inconsistencies frustrate attempts at linear reading or using the lexicon as a straightforward reference tool. These inconsistencies may reflect Ruland's source-oriented methodology. Rather than regularising terminology, he appears to preserve the orthography as it appears in his sources, thereby aligning his book with earlier Paracelsian glossaries, which were often appended to specific texts and served primarily as aids for readers of those particular works. Such books would list only the terms worth explaining found in the specific text, as they appear within that text. While such orthographic fidelity may have facilitated tracing sources for the book's users, it creates redundancies and obscures conceptual relationships between entries in Ruland's work that serves a different purpose. From a lexicographical standpoint, therefore, one might say that the reliance on orthography as the primary organising principle comes at the expense of systematisation.

⁴⁴ Soukup suggests that Ruland, who served as royal physician to Rudolf II, may not have held a high opinion of mercurius preparations. Rudolf Werner Soukup and Irmgard Soukup-Unterweger, "Terminologie alchemischer Technologie," in *Alchemie – Genealogie und Terminologie: Bilder, Techniken und Artefakte*, ed. Petra Feuerstein-Herz and Ute Frietsch (Wiesbaden: Harrassowitz, 2021), 91–102.

⁴⁵ "Alcohol" (C-entries), Ruland, *Lexicon Alchemiae*, 26–28; "Alcohol" (K-entries), Ruland, *Lexicon Alchemiae*, 30. Spelling variants such as *alkoel* and *alkol* are found only on page 30 (Figure 6), while C-spelled variants (e.g. *alcoool*) appear earlier, spanning pages 26–28 and amounting to nearly two full pages of content. Although these groups are internally clustered, they are separated across several pages and lack any system of cross-reference.

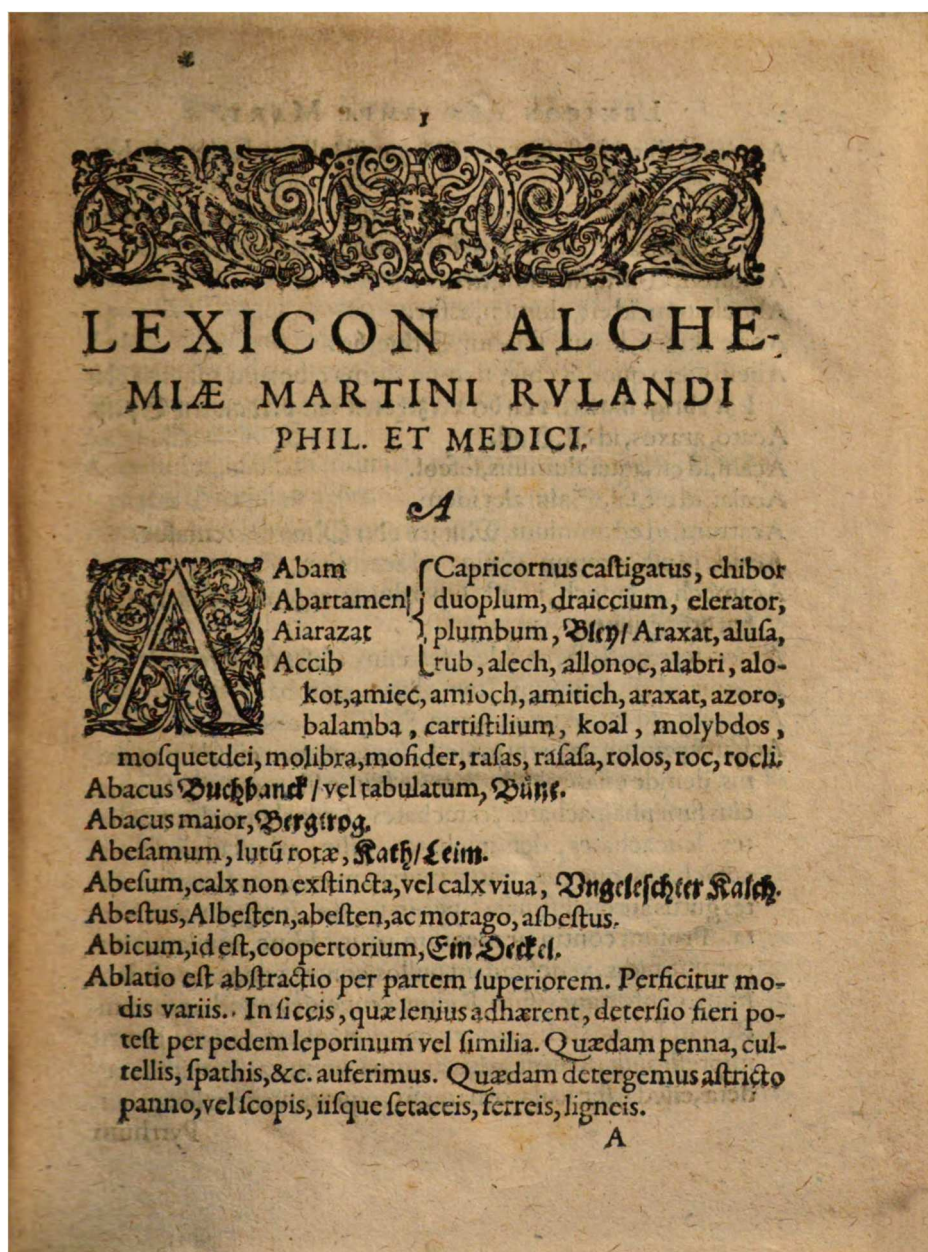


FIGURE 5 Beginning of Ruland's dictionary with the first page, containing the first (A-letter) entry, Augsburg, Staats- und Stadtbibliothek, 4 Phys 184, p. 1. Digital copy: BSB-ID 991037514909707356. <https://www.digitale-sammlungen.de/en/view/bsb11221307?page=8,9> (15.06.2026).

This approach of assigning separate headword status to what we would consider variants of a single word results in multiple entries for essentially the same term, differentiated only by minor orthographic variation. This physical and

typographical separation between what could be listed as variants of a single entry significantly impedes navigating the dictionary in ways modern readers would expect. A user consulting the K-spelled forms might not encounter the earlier C-spelled entries unless already aware of them. The K-variant entries are particularly brief, only two on page 30 (Figure 6), offering differing definitions depending on the precise spelling: *alkoel* is described as antimony, lead, or *lapis lazuli*, while the same term is elsewhere defined, in Paracelsian terms, as tartar or “the alkoel of wine,” a meaning introduced by Paracelsus himself and not used prior. By contrast, the C-spelled entries occupy nearly three pages and amount to approximately two full pages of content. One especially lengthy entry, *alcool*, nearly fills an entire page and centres on Paracelsian usages of the term. Ruland defines *alcohol* as *stibium* or antimony; *alcol* as vinegar (also giving the German term *Essig*).⁴⁶ Related to the meaning of *alcohol* as *pulvis*, *alcolismus* is defined as grinding down or corrosion, rather than alcohol abuse, as the modern German term now suggests. *Alkoel* he defines as refined lead from ore, noting that some say it is *lapis lazuli* (or, in German, *Lasurstein*), others antimony. *Alkol*, according to Ruland, is tartar; he cites Paracelsus, who states that *alkol* is the purer substance of a thing, separated from its impurity. Ruland thus concludes that the alcohol of wine is the most purified burning water (i.e. rectified spirit), the best, cleanest, subtlest, and most noble. The same explanation appears again, in much more detail, under the entry for *alcool*. Here, Ruland criticises those who interpret the term as powder most finely ground in iron or bronze pestles or mortars. A close reading of Paracelsus, he argues, clearly shows otherwise, as Paracelsus frequently discusses the alcohol of wine, meaning rectified burning water, which has nothing in common with powder. Therefore, Ruland concludes, the *alcool* of things is nothing other than the purer and cleaner substance separated from its impurity, stressing that the majority attempting this *arcanum* are deceived. This longer entry is likely to be Ruland’s own contribution, based on his close reading of Paracelsian texts, rather than material copied from elsewhere, though both methods of compilation are to be found across his entries. Again, it is important to note that these two largely similar entries, *alcool* and *alkol*, though differing in length and detail, would have appeared in different places in the dictionary. It remains unclear

⁴⁶ The following section is based on pages 26–28 (C-spellings) and page 30 (K-spellings) in the Latin original:

Ruland, *Lexicon Alchemiae*, 26–28: “*Alcohol* est pulvis subtilissimus. (..)

(*Alcool*) Nonnulli satis incongrue, sunt interpretati pro pul. minutissime trito, pistillo vel mortario ferreis, aeneisque. Sed error istorum satis arguitur eo, quod variis locis Paracels. de alcoole vini scribit, pro aqua ardente rectificata, quae quidem nihil cum puluere commune habet. Est igitur alcool rerum, aut corporum quorumcunque nihil aliud, quam purior & mundior substantia ab impura separata. (..) Maxima pars eorum, qui Paracelsi praeparationum anatomiam exercent rerum, tentantes arcanum hoc, fallitur. (..)

Alcolismus est comminutio vel corrosio. (..)

Alcol, id est, Acetum, *Essig*. (..)

Alcohol, id est, stybium siue antimonium. ”

Ruland, *Lexicon Alchemiae*, 30 and see also Figure 6: “*Alkoel*, id est, plumbum exquisitum de minera. *Etliche sagen es sey lapis lazuli, Lasurstein/etliche aber es sey Antimonium*.”

Alkol, est Tartarus. Theophrastus sic ait: *Alkol* est purior substantia rei, segregata ab impuritate sua. Sic *alkol* vini est aqua ardens rectificatae mundissima, *das beste und reineste/ das subtileste/ und edleste*.”

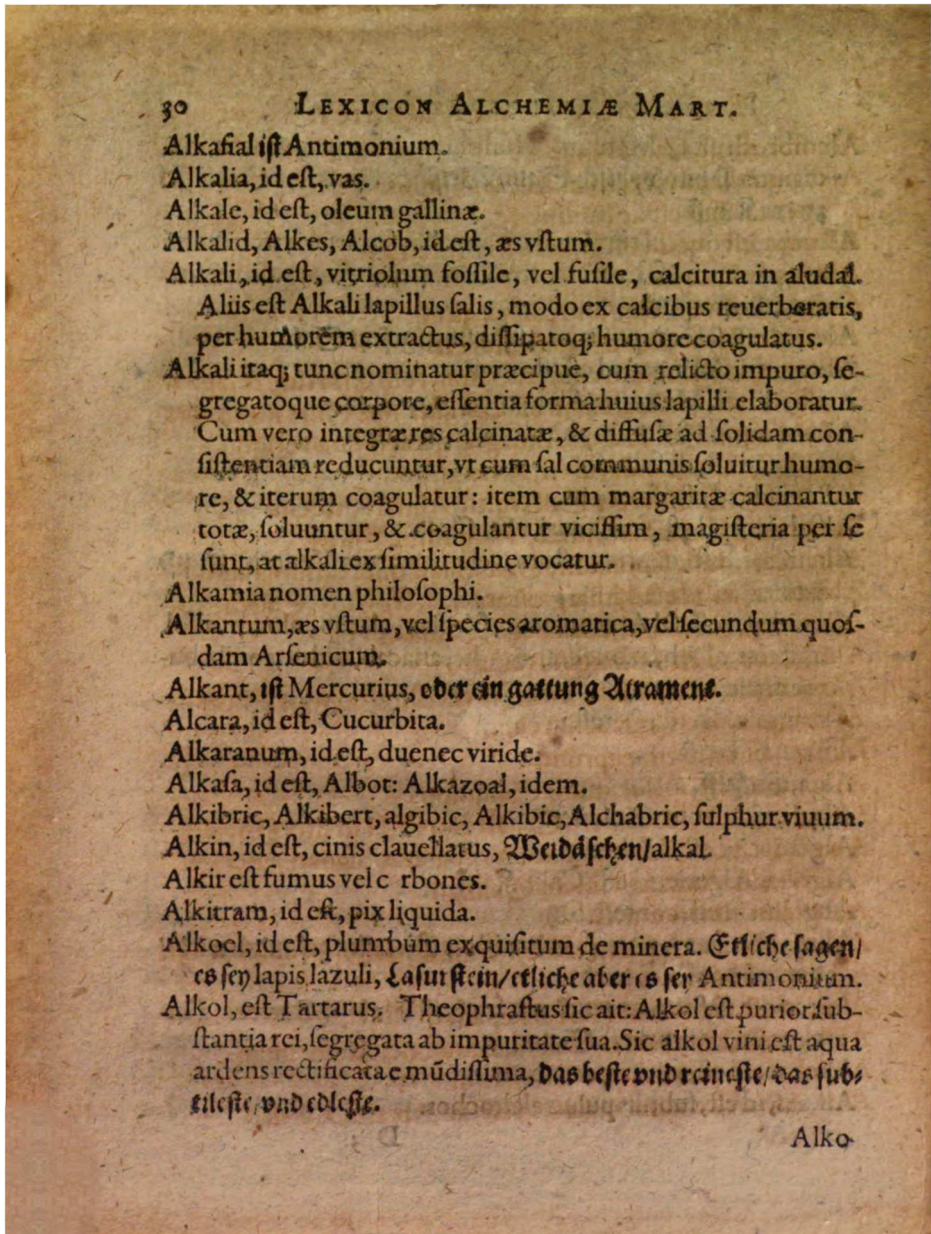


FIGURE 6 Page containing a range of entries relevant to this investigation, including “Alkali,” “Alkibric” and “Alkoel,” highlighting the different spellings in which terms appear and the lack of standardisation with multiple entries for the same term. Augsburg, Staats- und Stadtbibliothek, 4 Phys 184, p. 30. Digital copy: BSB-ID 991037514909707356. <https://www.digitale-sammlungen.de/en/view/bsb11221307?page=38,39> (15.06.2026).

whether Ruland deliberately placed his explanation in multiple locations to ensure it would be found, or whether he compiled the entries in a more unstructured manner and may not have recalled that he had already explained the term elsewhere. The fact

that he may not have had sufficient time to finalise the *Lexicon* due to his sudden death from the plague may be another factor in the matter. What remains is the fact that each entry offers a different referent and level of specificity, yet they are presented without cross-reference or clarification of their interrelationship. While some specific terms or meanings are accorded extended treatments, occupying several pages in some cases, others are only mentioned in passing. We conjecture that Ruland's inclusion of variant spellings, some of which are likely misspellings, may occur for the purpose of meeting the needs of a spectrum of readers with varying degrees of technical vocabularies in Latin, German, or Arabic.

Despite the limited overall number of entries for alcohol (compared to other terms in the dictionary), their semantic range is wide. No editorial framework is provided to guide readers through the overlapping or conflicting meanings. Instead, entries are arranged strictly according to orthography, with each spelling treated as a distinct lemma. This structural choice suggests that spelling, rather than meaning, functioned as Ruland's primary organisational principle. This arrangement reveals Ruland's methodological alignment with earlier Paracelsian *Onomastica* traditions. Using this model, originally intended for word lists appended to explain specialist terms in specific text, in a more general lexicon, the interpretation is delegated entirely to the reader, who is left to discern connections across variant forms and contexts. Ruland appears to have maintained the original spellings encountered in his source texts, preserving them in their discrete orthographic forms rather than consolidating them into unified entries. This explains the recurrence of multiple entries for semantically related terms and the absence of cross-referencing. While this might be seen as a lack of consideration for the reader by modern standards, it may have appeared advantageous to Ruland's contemporaries, because, unlike many similar resources at the time, this mode of presentation allowed the dictionary to be used outside of a constrained context, such as one specific book to which such a reference tool would be appended. This approach stands in marked tension with Ruland's stated aim in the preface, which was to bring clarity and order to the chaotic and obscure terminology of alchemy. In practice, the lexicon reproduces rather than resolves the linguistic ambiguity of the alchemical tradition. It reflects a descriptive rather than normative mode of lexicography, in which ambiguity is retained as a feature of the source material, not an error to be corrected. Ruland's treatment of *alcohol*, then, is emblematic of a broader editorial philosophy that foregrounds the multiplicity of alchemical language, leaving the task of reconciliation to the reader.

A similar semantic divergence appears under the headword *alumen*.⁴⁷ On a page containing many *alumen* entries, one explicitly describes a term as Arabic (*Alumen Calcilien est azub Arabice*, page 35, see Figure 7).⁴⁸ The text notes disagreement among authorities as to whether this substance is properly an alum or a type of

⁴⁷ The entries for *alumen* equally span several pages (33–36), suggesting its perceived significance with multiple orthographic and contextual variants, within the alchemical lexicon.

⁴⁸ "*Alumen*" (all entries), Ruland, *Lexicon Alchemiae*, 33–36.

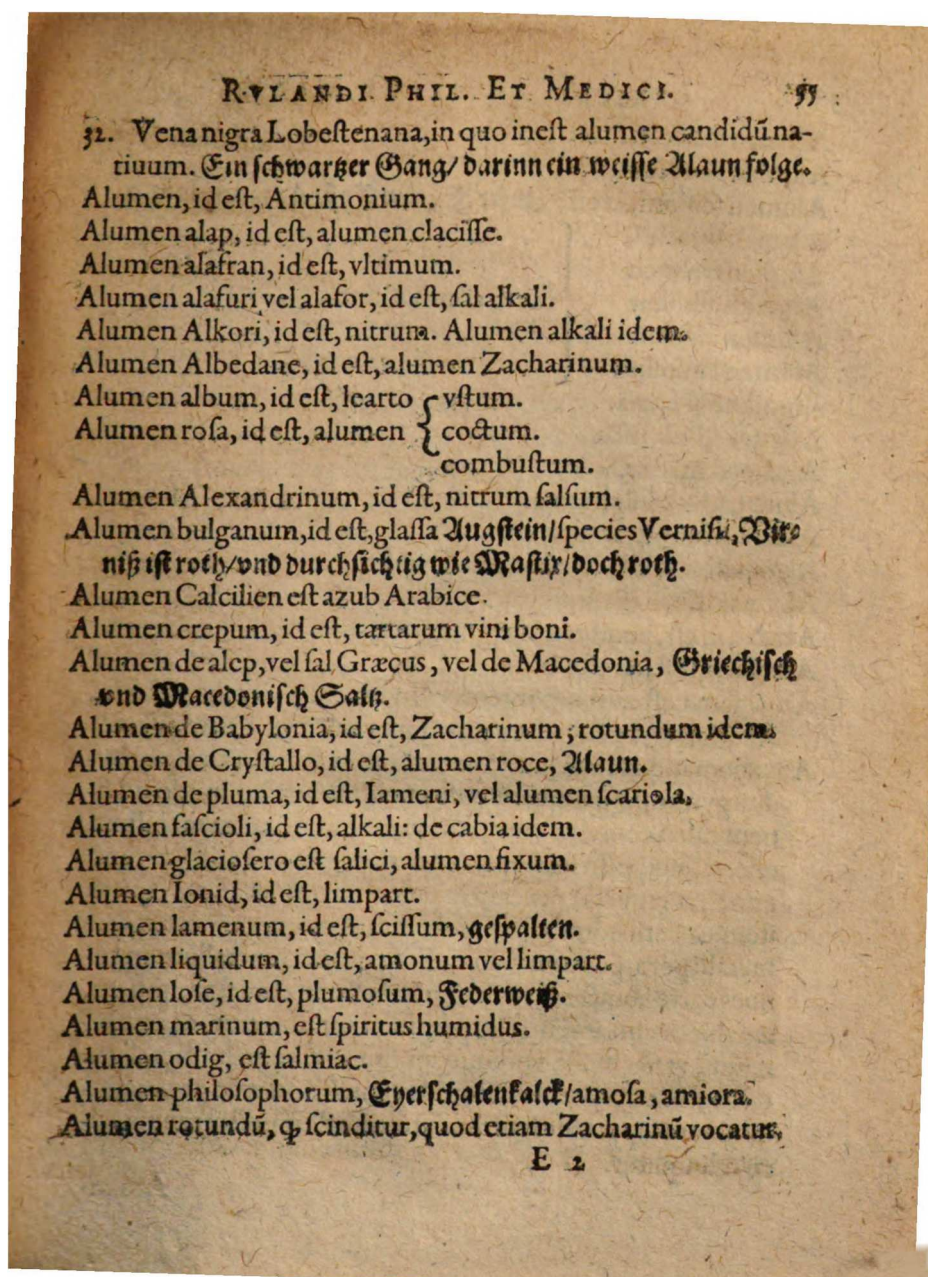


FIGURE 7 Ruland, *Lexicon Alchemiae*, page containing many “Alumen” entries, including one that explicitly describes a term as Arabic (*Alumen Calcilien est azub Arabice.*), Augsburg, Staats- und Stadtbibliothek, 4 Phys 184, p. 35. Digital copy: BSB-ID 991037514909707356. <https://www.digitale-sammlungen.de/en/view/bsb11221307?page=42,43> (15.06.2026).

alkali salt. As per usual, Ruland offers no reconciliation or overarching definition, simply multiple, partly contradictory perspectives listed adjacently. The aforementioned entry for *Sal*, for instance, extends across multiple pages and explicitly references both the Arabic *melech* and the Greek *ἄλας*, further exemplifying Ruland's approach which at times verges on the format of a multilingual dictionary.⁴⁹ Other entries display similar patterns. Terms derived from the Arabic *kibrīt* (sulphur), such as *Quebricum*, *Quebrit*, *Kibric*, and *Kibrith*, appear across several distinct entries, variously defined as sulphur, the philosophical "father of mercury" or even the stone itself.⁵⁰ Though the etymological relationship is evident, especially to a modern reader familiar with Arabic, Ruland treats some of them as independent terms, again offering no indication of their shared origin or semantic field. It is hard to tell whether he would have been aware of them or simply considered these terms distinct alchemical *Decknamen*. Yet, Ruland occasionally acknowledges orthographic variation, for example, in the case of *Quebricum*, by noting that the term "should be written with a K".

Taken together, these examples demonstrate Ruland's lexicographical method. While this practice maintains fidelity to the historical textual record, it produces a fragmented reference work in which closely related terms may appear disconnected, and the burden of interpretation falls entirely on the reader. Despite his stated goal in the preface, to bring order to alchemical terminology, the lexicon offers little in the way of editorial guidance. Rather than resolving this polysemy of alchemical language, Ruland reproduces its complexity, perhaps because he regarded such multiplicity not as a problem to be solved but as a reflection of alchemy's inherently polysemous character.

Mediating diverse knowledges in Ruland's Lexicon Alchemiae

As indicated in the introduction, reference works and other tools for organising knowledge became increasingly widespread in the sixteenth and seventeenth centuries.⁵¹ They developed into commercially successful genres that circulated broadly while also dominating specialised domains such as alchemy and chymistry. While recent scholarship has begun to reassess alchemical how-to literature, mining treatises, and other forms of technical writing, alchemical dictionaries have rarely been examined within these broader frameworks of knowledge organisation. One difficulty is the lack of direct evidence concerning how such dictionaries were compiled. Consequently, the present study has approached Martin Ruland's *Lexicon*

⁴⁹ "Sal" (all entries), Ruland, *Lexicon Alchemiae*, 410–21.

⁵⁰ Ruland, *Lexicon Alchemiae*, 400: "Quebricum, für *kebrit*. Etliche sagen / es sey Arsenicum / und gehöret zu schreiben mit dem K. Aber Stephanus sagt / es sey sulphur.

Quebrit. 1. sulphur. *quibrith* idem."

On p. 270: "Kibrith, id est, sulphur.

Kibric, ist deß Mercurii vnd aller flüssigen Ding Vatter / und die erste Materi darauß Mercurius wirdt, wirdt auch der lapis selbs genennet."

⁵¹ Ann M. Blair, *Too Much to Know: Managing Scholarly Information Before the Modern Age* (New Haven, CT: Yale University Press, 2010).

Alchemiae by working backwards from the surviving work itself in order to reconstruct its structure, compilation practices, and possible modes of use. Rather than undertaking a strictly lexicographical analysis, this article has examined the *Lexicon* as a knowledge-organisation resource. Through a combination of computational analysis and close reading, it has sought to illuminate the structural principles underlying Ruland's compilation. Quantitative analysis of the text through word counts and the identification of Arabic-derived vocabulary has made it possible to observe patterns within the *Lexicon* that are difficult to detect through traditional reading alone. The resulting dataset provides insights into the distribution and contextual usage of such terms, offering a structural perspective that complements earlier studies which have generally approached the *Lexicon* through investigating individual entries. While the present article has explored only part of this dataset, it demonstrates the potential of such methods for analysing early modern reference works and lays the groundwork for future, more detailed investigations that would have been beyond the scope of the current article. For instance, the presence of Hebrew or pseudo-Hebraic terms, or the treatment of vernacular translation within the dictionary, could be fruitfully examined as additional case studies using a similar methodology. We expect significant differences between the use of European vernaculars, representing local and practical forms of knowledge, and less accessible languages that were regarded as exotic and held particular appeal for many alchemists of the period as potential sources of occult knowledge.

The present analysis has been able to identify important insights into how Ruland's work relates to earlier traditions of Paracelsian lexicography. Earlier lexical aids, including short glossaries embedded within larger works, often consisted of very brief entries that merely equated one term with another. Such materials could accommodate numerous specialised terms in relatively few pages because the entries rarely extended beyond simple equivalences, such as, for instance, "aseb is alumen."⁵² Ruland's *Lexicon* differs markedly from these precedents. Even where it reproduces terms found in earlier lists, the entries are often expanded and accompanied by explanations. The work also contains a far greater number of entries overall, many of them considerably longer and more detailed. Both the scale of the compilation and the explanatory character of the entries distinguish it from most earlier lexical aids and may help explain its success as a standalone reference tool.

Seen in this light, the *Lexicon Alchemiae* represents both the culmination and the transformation of Paracelsian lexicographical traditions. Building upon Gerhard Dorn's *Dictionarium Theophrasti Paracelsi* (1583), Ruland incorporated large portions of earlier material while simultaneously expanding and reinterpreting it.⁵³ As

⁵² Hieronymus Reusner, "Synonyma," in *Pandora, das ist, die edleste Gab Gottes* (Basel: Samuel Apiarius, 1582), 253–309 (on 267), <https://digital.sciencehistory.org/works/d1veeyn/viewer/3p5bkn> (accessed 22 February 2026).

⁵³ Dorn, *Dictionarium Theophrasti Paracelsi*.

Kühlmann has argued, the significance of Ruland's intervention lies less in producing a merely more complete resource than in recognising that inherited alchemical terminology had to be reconciled with contemporary experience.⁵⁴ The inclusion of entries such as *lapis*, which do not belong exclusively to Paracelsian vocabulary, illustrates how the lexicon extends beyond a strictly Paracelsian framework. Through the incorporation of terms from medicine, metallurgy, and other related fields, Ruland transformed earlier Paracelsian onomastica into a broadly conceived, alphabetically organised lexicon of alchemical *realia*. In this respect the work belongs not only to the history of lexicography but also to the wider tradition of early modern handbooks of practical knowledge.

The multilingual character of the lexicon further reflects the intellectual conditions in which it emerged. Ruland writes for readers who encountered alchemical, medical, and metallurgical texts in Latin as well as in various vernacular languages. His entries therefore move fluidly between Latin and German, while also incorporating loanwords from Greek, Hebrew, and Arabic. Many of these terms probably did not refer to texts directly read by his audience; rather, they circulated as technical expressions or as *Decknamen* within alchemical literature. Ruland stands out as a particularly prominent practitioner of bilingual explanation in Latin and German, although he employs this strategy unsystematically, alternating freely between the two languages. The result resembles the linguistic situation observed by Dominik Berrens in Georg Agricola's (1494–1555) *Bermannus*, where one's ability to comprehend the Latin technical terms presupposes a solid knowledge of German and familiarity with multiple terminological traditions.⁵⁵ Such linguistic hybridity reflects broader dynamics within early modern knowledge cultures. Recent scholarship has emphasised the inherent multilinguality of early modern scholarship.⁵⁶ Writers and readers routinely worked across several languages, and part of their intellectual competence lay in navigating and connecting these linguistic domains. Within this context, alchemical dictionaries can be understood not merely as pedagogical aids but as instruments for negotiating a multilingual technical vocabulary. They intersect with humanist philology, projects of universal language, and even contemporary developments in cryptography, all of which are likewise concerned with the systematisation and interpretation of different forms of language.⁵⁷

At the same time, the period witnessed a growing integration of practical, craft-based knowledge into scholarly discourse, particularly in fields related to medicine, chemistry, and mining.⁵⁸ Technical expertise that had long circulated in artisanal contexts increasingly entered printed literature and scholarly debate. This convergence of different domains of knowledge created both the conditions and the

⁵⁴ Kühlmann, "Materia medica als Problem," 160–64.

⁵⁵ Berrens, "Names and Things."

⁵⁶ Most, Schäfer, and Söderblom Saarela, *Plurilingualism in Traditional Eurasian Scholarship*.

⁵⁷ Soukup and Soukup-Unterweger, "Terminologie alchemischer Technologie"; Hannah Marcus and Paula Findlen, "Deciphering Galileo: Communication and Secrecy Before and After the Trial," *Renaissance Quarterly* 72, no. 3 (2019): 953–95; Anthony Grafton, *Defenders of the Text: The Traditions of Scholarship in an Age of Science, 1450–1800* (Cambridge, MA: Harvard University Press, 2003).

⁵⁸ Smith, *From Lived Experience to the Written Word*.

demand for new forms of reference works capable of bringing together diverse materials. Ruland's *Lexicon* reflects precisely such a convergence. Although rooted in Paracelsian philology and in the extraction of terminology from existing texts, it also incorporates material derived from adjacent practical disciplines and perhaps even from oral informants, as Kühlmann suggests.⁵⁹ In this sense the lexicon belongs to the broader early modern culture of compilation that Ann Blair has described as a response to the problem of information overload.⁶⁰ Within this culture of compilation, reference works functioned as shared scholarly tools, not only as authoritative repositories of information but as reading notes that others might have wanted to create for themselves. Ruland's *Lexicon* fits this description well. Rather than presenting a fully coherent or standardised system, it resembles a published collection of working notes assembled from reading, translation, and comparison across multiple sources. Its dependence on earlier lexica, especially Dorn's *Dictionarium*, is evident, yet Ruland frequently expands upon his predecessors or diverges from their interpretations, demonstrating editorial agency within the process of compilation. The lexicon's contemporary usefulness is suggested by the ten-year imperial privilege granted to its first edition, as well as by later reprints in Frankfurt (1661) and Nuremberg (1671) and its English adaptation by William Johnson as the *Lexicon Chymicum* (1652).⁶¹

The conceptual character of the work may be illuminated through recent historiographical approaches to hybrid forms of knowledge. In his study of Hieronymus Brunschwig (ca. 1450–ca. 1512), Tillmann Taape has described similar phenomena through the concept of “braided knowledge,” drawing on earlier notions such as hybrid knowledge and trading zones.⁶² Crucially, Taape demonstrates that such categories are not merely retrospective analytical tools: early modern actors themselves actively negotiated the integration of diverse linguistic and epistemic traditions within their writings. However, Ruland's work goes beyond linguistic hybridity alone. In a manner similar to Brunschwig, he braids together knowledge from different intellectual spheres, bringing together elements from several

⁵⁹ Kühlmann, “Materia medica als Problem,” 160–64.

⁶⁰ Blair, *Too Much to Know*.

⁶¹ William Johnson, *Lexicon Chymicum* (London: William Nealand, 1652); Weimann, “Paracelsus-Lexikographie in vier Jahrhunderten,” 180–81.

⁶² Taape, *Crafting Medicine*, see esp. 3–15 and 233–39. On the concept of “trading zones” (a term coined by Peter Galison in 1997) see Peter Galison, *Image and Logic: A Material Culture of Microphysics* (Chicago, IL: University of Chicago Press, 1997), 781–844; Derren Rosbach, “Building a Transdisciplinary Trading Zone,” *International Journal of Science in Society* 3, no. 3 (2012): 17–29, <https://doi.org/10.18848/1836-6236/CGP/v03i03/51334>; Michael E. Gorman, “Levels of Expertise and Trading Zones: A Framework for Multidisciplinary Collaboration,” *Social Studies of Science* 32, no. 5–6 (2002): 933–38, <http://www.jstor.org/stable/3183060>. On hybrid knowledge, see for example: Anna Winterbottom, *Hybrid Knowledge in the Early East India Company World*, (London: Palgrave Macmillan, 2016), 1–25. On braided knowledge see: Robin Wall Kimmerer, *Braiding Sweetgrass: Indigenous Wisdom, Scientific Knowledge and the Teachings of Plants* (Minneapolis, MN: Milkweed Editions, 2013); Projit Bihari Mukharji, *Doctoring Traditions: Ayurveda, Small Technologies, and Braided Sciences*, (Chicago, IL: University of Chicago Press, 2016), 1–34, <https://doi.org/10.7208/9780226381824-002>; Projit Bihari Mukharji, “Casting Blood Circulations: Translatability and Braiding Sciences in Colonial Bengal,” in “Translating Medicine across Pre-modern Worlds,” ed. Tara Alberts, Sietske Fransen and Elaine Leong, *Osiris* 37 (2022): 192–210, <https://doi.org/10.1086/719226>.

traditions and attempts to organise them within a single reference work. The result is a compilation that reflects the convergence of different bodies of knowledge while simultaneously striving to systematise them. Linguistically, it braids together Latin, German, and loanwords from Greek, Hebrew, and Arabic; conceptually, it intertwines Paracelsian terminology with broader alchemical, medical, and metallurgical vocabularies. The work thus embodies the interaction between learned and vernacular traditions in what Taape calls “vernacular text work.”⁶³

Seen from this perspective, the *Lexicon Alchemiae* occupies a transitional position within early modern knowledge organisation. It consolidates earlier Paracelsian lexicographical efforts while simultaneously expanding them into a more comprehensive alchemical reference work. Yet its structure also reveals the limits of systematisation. Although Ruland announces in his preface an intention to resolve the confusion of alchemical terminology, the lexicon ultimately preserves much of the instability and semantic multiplicity characteristic of the field. Cross-referencing remains inconsistent, synonymous terms are not always consolidated, and morphologically related entries are frequently treated separately. These features suggest that to Ruland, the plurality of alchemical language was not simply a problem to be eliminated but a structural condition of the discipline itself. Ruland’s work, thus, demonstrates both the ambition to organise such material systematically and the practical limits of that ambition. As can be seen from this, and other studies of alchemical glossary traditions, the systematisation of alchemical language was not achieved once and for all in a single reference work. Rather, it advanced incrementally, with each new compilation building upon and extending the efforts of its predecessors.

The structure of the dictionary reveals a keen awareness of earlier lexicographical models and of the features that had proven attractive to readers. Ruland appears to have incorporated several elements that had contributed to the success of earlier Paracelsian lexica, including multilingual terminology and the transformation of word lists from appendices to independent reference works. At the same time, he expanded the scope of the lexicon beyond strictly Paracelsian vocabulary and by integrating more general medical and scientific terminology, the work extended its relevance to a broader alchemical and medico-chemical readership. In doing so, it blurred earlier distinctions between specialised Paracelsian terminology and more widely used technical language, contributing to the diffusion of formerly specialised terms into a broader vocabulary. In its inconsistencies and seeming lack of methodological transparency, Ruland’s *Lexicon Alchemiae* offers valuable insight into how early modern scholars navigated multilingual, multi-sourced, and conceptually fluid domains of knowledge.

By combining computational analysis with close reading, the present study has attempted to illuminate these dynamics and to situate the *Lexicon Alchemiae* within broader early modern practices of compilation and knowledge organisation.

⁶³ Taape, *Crafting Medicine*, 15.

The dataset generated through this analysis, particularly the material concerning Arabic-derived terminology, opens further avenues for research that will allow the linguistic and conceptual networks embedded in Ruland's book to be examined in greater detail in future work. Together, such approaches may contribute to a more comprehensive understanding of alchemical dictionaries as significant instruments in the early modern management and transmission of diverse knowledges.

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Statement on the Use of AI

During the preparation of this manuscript, the authors have used ChatGPT 4o for proofreading and improving the English language. The authors have reviewed and edited the output and take full responsibility for the content of this publication.

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